Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

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Dataset at a Glance

3,900	18	37
Purchases	Columns	Missing Ratings

Key Features

- **Demographics:** Age, Gender, Location, Subscription
- **Purchase:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Ratings, Shipping

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Data Preparation in Python

1

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

2

Clean & Impute

Filled missing Review Ratings with median per product category

3

Feature Engineering

Created `age_group` and `purchase_frequency_days` columns

4

Load to PostgreSQL

Connected Python to PostgreSQL; dropped redundant `promo_code_used`

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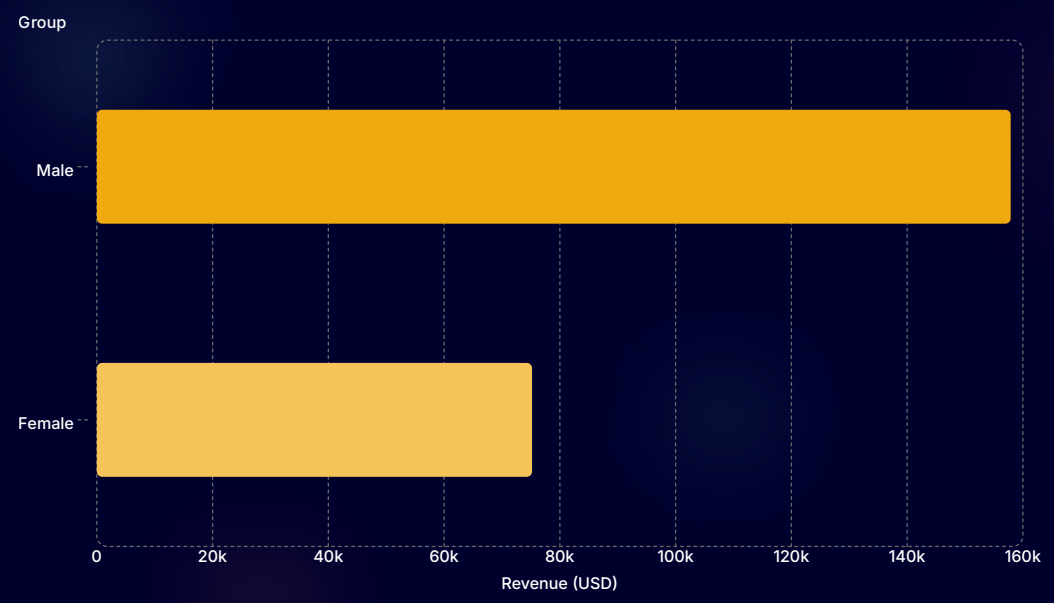
Revenue & Spending Insights



Shipping Type vs. Avg. Spend

Standard

$58.46 avg. purchase

Express

$60.48 avg. purchase

839 high-spending customers used discounts yet spent **above average**.

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Top Products & Discount Dependency

Top 5 by Rating

Gloves
⭐ 3.86

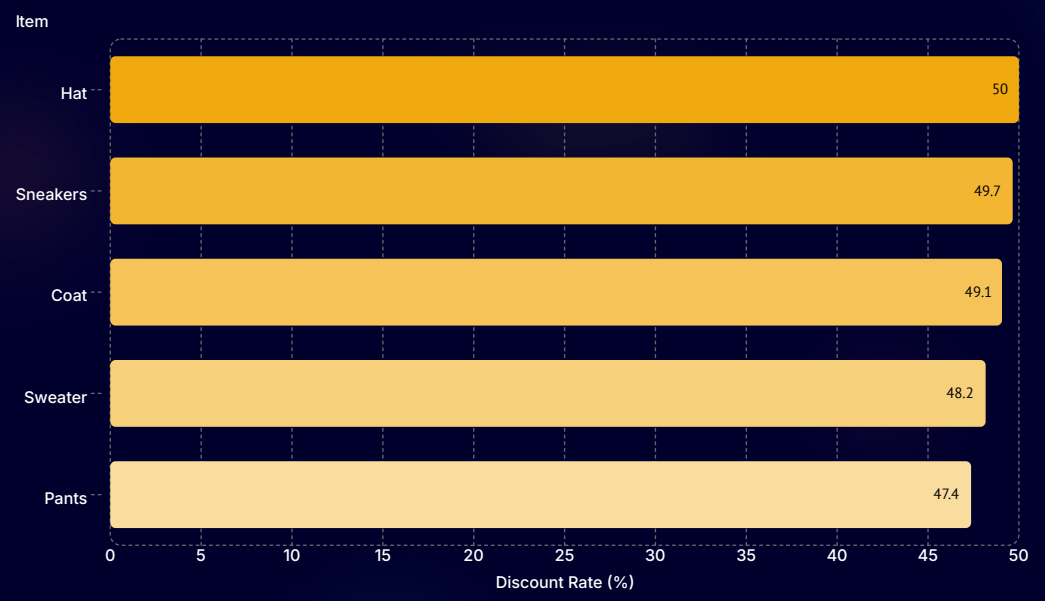
Sandals
⭐ 3.84

Boots
⭐ 3.82

Hat
⭐ 3.80

Skirt
⭐ 3.78

Most Discount-Dependent Products



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Customer Segmentation & Subscriptions

three tiers showing purchase history classification

Subscribers vs. Non-Subscribers

Status	Customers	Avg Spend	Revenue
Yes	1,053	$59.49	$62,645
No	2,847	$59.87	$170,436

Repeat buyers (>5 purchases): **958 subscribed** vs. 2,518 non-subscribed.

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Top Products by Category

👗 Clothing

- Blouse – 171
- Pants – 171
- Shirt – 169

💍 Accessories

- Jewelry – 171
- Sunglasses – 161
- Belt – 161

👟 Footwear

- Sandals – 160
- Shoes – 150
- Sneakers – 145

🧥 Outerwear

- Jacket – 163
- Coat – 161

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Power BI Dashboard

Key KPIs

3.9K

Customers

$59.76

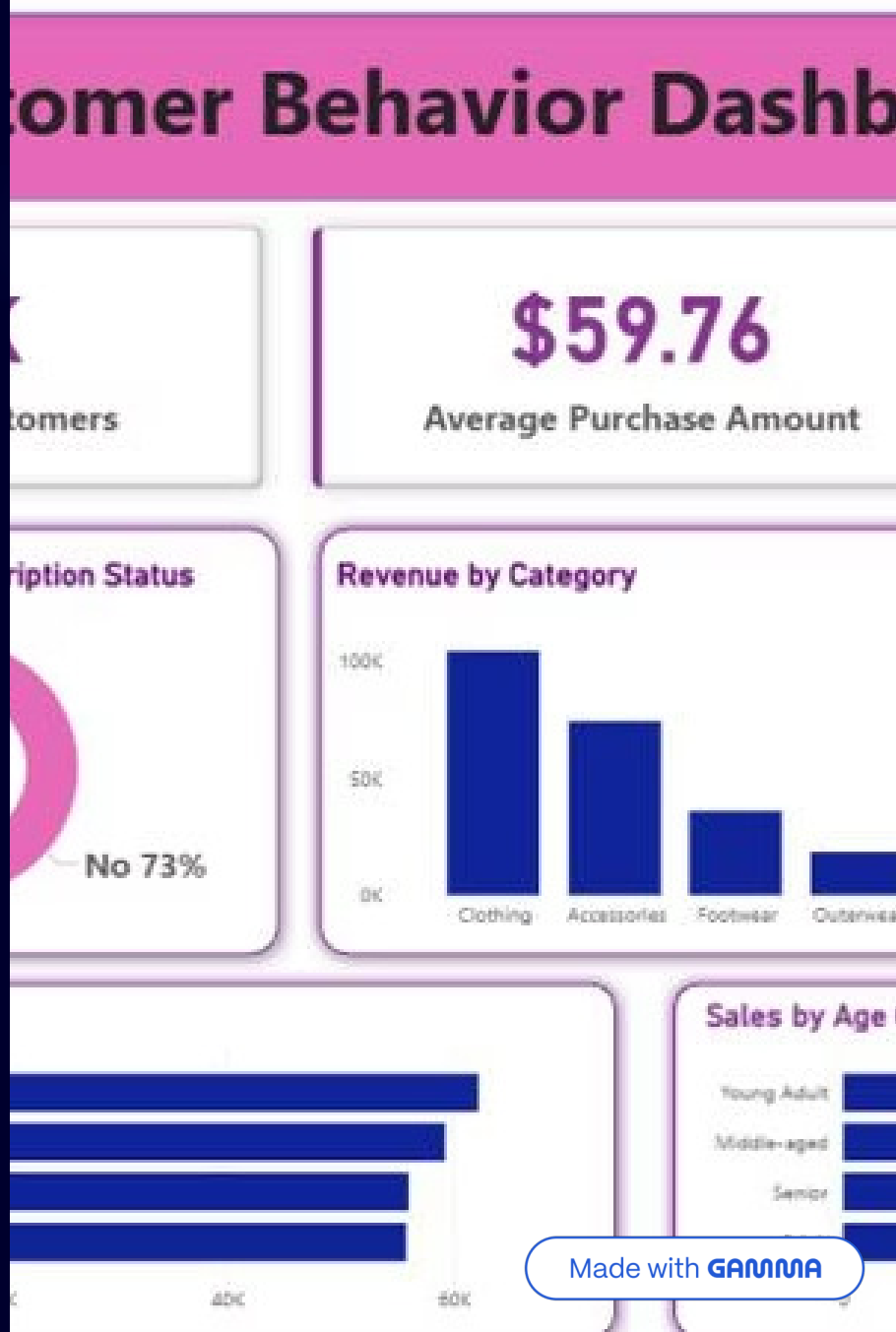
Avg. Purchase

3.75

Avg. Rating

27% subscribers · **73%** non-subscribers

Clothing leads revenue (~$100K), followed by Accessories (~$75K).



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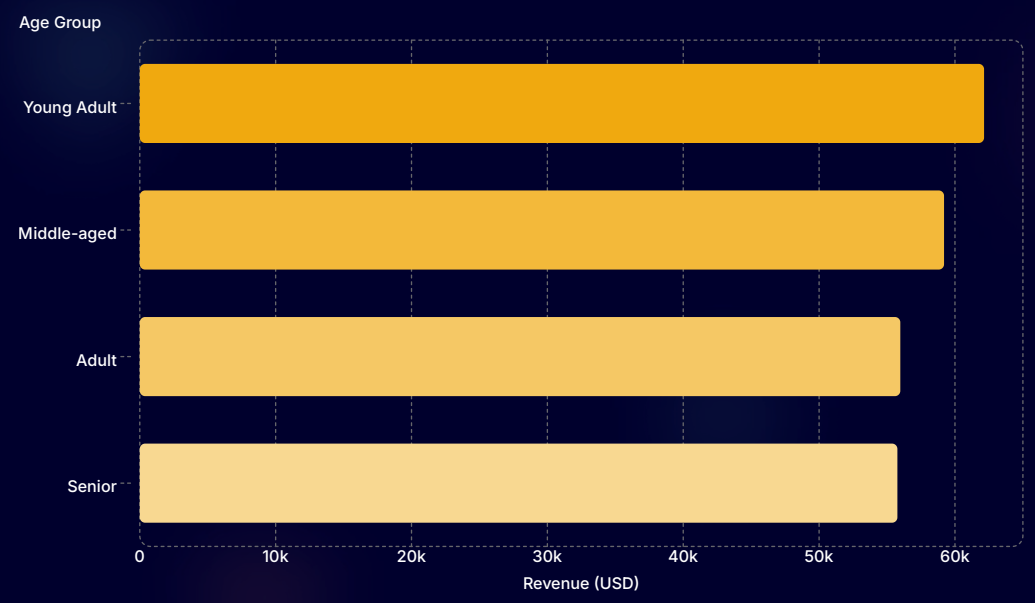
Revenue by Age Group



Young Adults Lead Revenue

Young Adults contribute the most at **$62,143**, with relatively balanced spend across all age groups.

All age groups fall within a narrow $6K revenue band — no single group dominates overwhelmingly.

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscribers

Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment

Review Discount Policy

Balance sales boosts with margin control on high-discount products

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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